

Mixed problems (2/3)

10.4–10.6

Outline

Christmas trees and boxes of lights

Shipments in boxes

Summer camps

Christmas trees and boxes of lights

Introduction: Trees and lights

- ▶ A problem with **two storeys**: below the trees to decorate, above the boxes of lights to buy
- ▶ The two storeys are tied by an **availability** constraint, colour by colour: no big-M
- ▶ We will see the link $z_c \leq x_c$ between a configuration and its indicator, and why no big-M is needed here
- ▶ We will build a primal bound with a two-phase constructive heuristic and an almost exact dual bound by hand

Problem 10.4: trees and lights (1/2)

- ▶ The city administration wants to decorate $q \in \mathbb{Z}_{\geq 1}$ trees for the holidays
- ▶ Every tree can be decorated according to one of the $n \in \mathbb{Z}_{\geq 1}$ possible configurations, each characterised by lights of $m \in \mathbb{Z}_{\geq 1}$ different colours
- ▶ For every configuration $c \in \{1, \dots, n\}$ and every colour $l \in \{1, \dots, m\}$, $u_{cl} \in \mathbb{Z}_{\geq 0}$ is the number of lights of colour l required by configuration c , and $i_c \in \mathbb{Q}_{>0}$ is the installation cost of a tree decorated that way

Problem 10.4: trees and lights (2/2)

- ▶ All the lights must be bought on the market, where they are sold in boxes of $k \in \mathbb{Z}_{\geq 1}$ types
- ▶ For every type $b \in \{1, \dots, k\}$ and every colour l , $v_{bl} \in \mathbb{Z}_{\geq 0}$ is the number of lights of colour l contained in a box of type b , and $p_b \in \mathbb{Q}_{\geq 0}$ is its cost
- ▶ To ensure a pleasant visual variety at least $f \in \mathbb{Z}_{\geq 1}$ different configurations must be used
- ▶ The goal is to decorate the trees at **minimum total cost**

The problem in words

- ▶ We **decide** how many trees to decorate with each configuration and how many boxes to buy of each type
- ▶ **The objective:** minimum total cost (installation plus boxes)
- ▶ **The constraints:**
 - ▶ all the trees must be decorated
 - ▶ the lights bought must be enough for those required, colour by colour
 - ▶ at least f different configurations

Decision variables

n integer variables and k integer variables:

x_c = trees decorated with configuration c , y_b = boxes bought of type

n binary variables:

$$z_c = \begin{cases} 1 & \text{if configuration } c \text{ is used} \\ 0 & \text{otherwise} \end{cases}$$

Model 10.4 (1/2)

Definition 1: Model 10.4 — trees and lights: objective, trees and lights

$$\min \quad \sum_{c=1}^n i_c x_c + \sum_{b=1}^k p_b y_b$$

$$\text{s.t.} \quad \sum_{c=1}^n x_c = q$$

$$\sum_{b=1}^k v_{bl} y_b - \sum_{c=1}^n u_{cl} x_c \geq 0, \quad \forall l \in \{1, \dots, m\}$$

Definition 2: Model 10.4 — trees and lights: variety, link and domains

$$\sum_{c=1}^n z_c \geq f$$

$$x_c - z_c \geq 0, \quad \forall c \in \{1, \dots, n\}$$

$$x_c \in \mathbb{Z}_{\geq 0}, \quad \forall c \in \{1, \dots, n\}$$

$$y_b \in \mathbb{Z}_{\geq 0}, \quad \forall b \in \{1, \dots, k\}$$

$$z_c \in \{0, 1\}, \quad \forall c \in \{1, \dots, n\}$$

Model description

- ▶ The **objective function** adds up the cost of the installations and that of the boxes bought
- ▶ The **trees** constraint, a single one, says that the decorated trees are exactly q
- ▶ The **lights** constraints, one per colour, say that the lights bought are enough to cover those used
- ▶ The **variety** constraint, a single one, imposes at least f different configurations
- ▶ The **link** constraints, one per configuration, tie the indicator z_c to the actual use of the configuration

The link between the two storeys

- ▶ The lights constraint is the **availability**: for every colour, the lights bought must be at least those required
- ▶ There is no **big-M**, because both sides are sums of real quantities
- ▶ The count of the boxes comes from here: if the lights of colour l required are u_l and a box contains v_{bl} of them, buying only boxes of type b one would need $\lceil u_l/v_{bl} \rceil$ boxes
- ▶ The model does not write the ceiling: it obtains it by making y_b **integer**

The “configuration used” link

The link constraint reads $z_c \leq x_c$.

- ▶ If $z_c = 1$ then $x_c \geq 1$: the configuration is really used on at least one tree
- ▶ The other direction is not imposed, but it is not needed: the minimisation objective has no interest in raising z_c , and variety forces it to do so exactly f times
- ▶ No big-M is needed here because x_c and z_c are directly comparable: x_c is a **count**, not a continuous quantity
- ▶ A frequent mistake: writing $x_c \leq M z_c$ thinking of activation. That would be the link in the *other* direction, true but useless, because it is already guaranteed by $\sum_{c=1}^n x_c = q$ and by $x_c \geq 0$

The model in gurobipy (1/2)

```
m = gp.Model("lights")
x = m.addVars(nc, vtype=GRB.INTEGER, name="x")
y = m.addVars(nb, vtype=GRB.INTEGER, name="y")
z = m.addVars(nc, vtype=GRB.BINARY, name="z")
m.setObjective(
    gp.quicksum(i[c] * x[c] for c in range(nc))
    + gp.quicksum(p[b] * y[b] for b in range(nb)),
    GRB.MINIMIZE)
```

The model in gurobipy (2/2)

```
m.addConstr(x.sum() == q, name="trees")
m.addConstrs((gp.quicksum(v[b][l] * y[b]
                           for b in range(nb))
               - gp.quicksum(u[c][l] * x[c]
                           for c in range(nc)) >= 0
               for l in range(nl)), name="lights")
m.addConstr(z.sum() >= f, name="variety")
m.addConstrs((x[c] - z[c] >= 0 for c in range(nc)),
              name="used")
```

The instance

$q = 20$ trees, $n = 3$ configurations, $m = 2$ colours, $k = 2$ box types, $f = 2$:

u_{cl}	colour 1	colour 2	i_c
$c = 1$	4	2	7
$c = 2$	2	3	6
$c = 3$	2	2	8

v_{bl}	colour 1	colour 2	p_b
$b = 1$	10	2	100
$b = 2$	15	4	200

The price of one light depends on the colour and on the box type: colour 1 at 10 in box 1 and at 40/3 in box 2; colour 2 at 50 in both. Colour 2 is much more expensive, and it is what drives the solution.

Model 10.4 on the instance (1/2)

Objective, the trees constraint and the two availability constraints on the lights:

$$\min \quad 7x_1 + 6x_2 + 8x_3 + 100y_1 + 200y_2$$

$$\text{s.t.} \quad x_1 + x_2 + x_3 = 20$$

$$-4x_1 - 2x_2 - 2x_3 + 10y_1 + 15y_2 \geq 0$$

$$-2x_1 - 3x_2 - 2x_3 + 2y_1 + 4y_2 \geq 0$$

Model 10.4 on the instance (2/2)

Variety ($f = 2$), link and domains:

$$z_1 + z_2 + z_3 \geq 2$$

$$x_1 - z_1 \geq 0$$

$$x_2 - z_2 \geq 0$$

$$x_3 - z_3 \geq 0$$

$$x_1, x_2, x_3 \in \mathbb{Z}_{\geq 0}$$

$$y_1, y_2 \in \mathbb{Z}_{\geq 0}$$

$$z_1, z_2, z_3 \in \{0, 1\}$$

Constructive heuristic: the primal bound (1/2)

Two phases.

- ▶ **Configurations:** $q - f + 1$ trees with the one that is cheapest to install and one tree for each of the other $f - 1$, so that variety is met at the lowest installation cost
- ▶ **Boxes:** while some light is missing, buy the box with the lowest price per missing light

It is a constructive heuristic: a single solution, one element at a time, never backtracking.

Constructive heuristic: the primal bound (2/2)

On the instance the cheapest configuration to install is number 2 (cost 6), the second one is number 1 (cost 7): 19 trees are decorated with the 2 and one with the 1.

- ▶ Then 42 lights of colour 1 and 59 of colour 2 are needed
- ▶ The heuristic buys 30 boxes of type 1

$$z(MILP) \leq UB = 3121$$

The heuristic picks the configuration that is cheapest to install, number 2, which is however the greediest for the expensive colour: the boxes pay the bill. It is the typical mistake of a constructive heuristic that looks at a single cost item.

Dual of the LP relaxation of model 10.4

α free with the trees (an equality), $\beta_l \geq 0$ with the lights, $\gamma \geq 0$ with the variety, $\delta_c \geq 0$ with the link.

Definition 3: Dual of the LP relaxation of model 10.4

$$\max \quad q\alpha + f\gamma$$

$$\text{s.t.} \quad \alpha - \sum_{l=1}^m u_{cl} \beta_l + \delta_c \leq i_c, \quad \forall c \in \{1, \dots, n\}$$

$$\sum_{l=1}^m v_{bl} \beta_l \leq p_b, \quad \forall b \in \{1, \dots, k\}$$

$$\gamma - \delta_c \leq 0, \quad \forall c \in \{1, \dots, n\}$$

$$\alpha \geq 0, \quad \beta_l \geq 0, \quad \gamma \geq 0, \quad \delta_c \geq 0$$

Dual on the instance (1/2)

Objective, the three columns of the x_c and the two columns of the y_b :

$$\begin{aligned} \max \quad & 20\alpha + 2\gamma \\ \text{s.t.} \quad & \alpha - 4\beta_1 - 2\beta_2 + \delta_1 \leq 7 \\ & \alpha - 2\beta_1 - 3\beta_2 + \delta_2 \leq 6 \\ & \alpha - 2\beta_1 - 2\beta_2 + \delta_3 \leq 8 \\ & 10\beta_1 + 2\beta_2 \leq 100 \\ & 15\beta_1 + 4\beta_2 \leq 200 \end{aligned}$$

Dual on the instance (2/2)

The three columns of the z_c and the domains:

$$\gamma - \delta_1 \leq 0$$

$$\gamma - \delta_2 \leq 0$$

$$\gamma - \delta_3 \leq 0$$

$$\alpha \stackrel{\geq}{\leq} 0$$

$$\beta_1, \beta_2 \geq 0$$

$$\gamma \geq 0$$

$$\delta_1, \delta_2, \delta_3 \geq 0$$

The variable α is free because the trees constraint is an equality.

Dual description

- ▶ α is the value of one decorated tree, β_l the price of one light of colour l , γ the price of variety, δ_c that of the link between configuration c and its indicator
- ▶ The **objective** prices the q trees at α and the threshold f at γ
- ▶ **Columns of the x_c :** using configuration c on a tree is worth α , consumes u_{cl} lights of each colour and releases δ_c ; the balance cannot exceed the installation cost i_c
- ▶ **Columns of the y_b :** a box of type b supplies v_{bl} lights of each colour, and their total value cannot exceed the price p_b
- ▶ **Columns of the z_c :** the price of variety cannot exceed that of the link which makes it claimable

A dual solution by hand (1/2)

- ▶ Set $\gamma = 0$ and $\delta_c = 0$: variety is not priced and the columns of the z_c are satisfied
- ▶ What is left are the columns of the y_b , which cap β , and those of the x_c , which cap α

One prices *a single* colour, at the highest price per light that no box can beat, and then takes the largest feasible α :

$$\bar{\beta}_l = \min_{b: v_{bl} > 0} \frac{p_b}{v_{bl}}, \quad \bar{\alpha} = \min_c (i_c + u_{cl} \bar{\beta}_l)$$

Every tree costs at least the installation of the most convenient configuration, lights included.

A dual solution by hand (2/2)

On the instance the better colour is number 2, where both box types give the same price per light, $\bar{\beta}_2 = 50$:

$$i_1 + 2 \cdot 50 = 107, \quad i_2 + 3 \cdot 50 = 156, \quad i_3 + 2 \cdot 50 = 108$$

hence $\bar{\alpha} = 107$ and

$$z(MILP) \geq LB = 20 \cdot 107 = 2140$$

Optimal solution

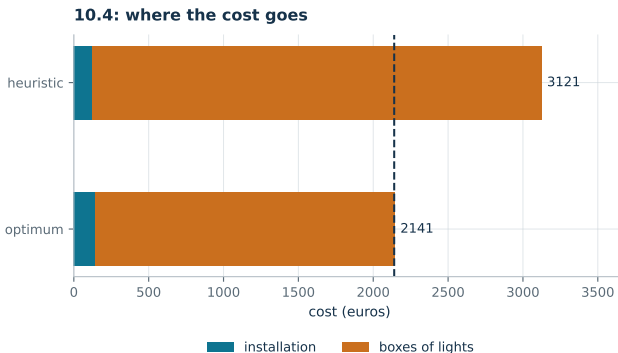
The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there. 19 trees are decorated with configuration 1 and one with configuration 3, and 20 boxes of type 1 are bought:

$$z(MILP) = 19 \cdot 7 + 8 + 20 \cdot 100 = 2141$$

The lights needed are 78 of colour 1 (200 are bought: many are left over) and 40 of colour 2 (exactly 40 are bought).

$$2140 = LB = z(LP) \leq z(LP^+) = 2141 = z(MILP) \leq UB = 3121$$

The optimal solution in a picture



Where the cost goes in the two solutions: installation and boxes, with the dual bound dashed. The dual bound is off by *one* unit out of 2141, the strengthened relaxation is exact, and the heuristic gap is 45.8%.

Additional remarks

- ▶ Colour 1 is left over in abundance: the boxes are bought for colour 2, and colour 1 comes “for free”. That is why the dual bound pricing colour 2 alone is almost exact
- ▶ The trees constraint is an equality. If it were $\leq q$ the problem would become trivial ($x = 0$ and zero cost); if it were $\geq q$ nothing would change, because decorating more trees costs more
- ▶ The structure “two storeys tied by an availability” appears identically in *cutting stock* problems: below the pieces to be cut, above the bars to be bought

Modelling question: minimum lot per configuration

Every configuration used must decorate at least three trees (below that threshold it is not worth equipping the crew).

How does the model change? What is the new optimum?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
UB	3121	heuristic solution
LB	2140	certified bound
$z(MILP)$	2141	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 45.8\%$
- ▶ Certified gap, known *before* solving the MILP (it cannot have $z(MILP)$ in the denominator): $UB - LB = 981$, that is $\frac{UB - LB}{|UB|} = 31.4\%$

Worked variant: all the configurations (1/2)

All three configurations are required to appear: only the datum f changes, from 2 to 3.

- ▶ The optimum rises from 2141 to 2239: forcing the use of configuration 2, the greediest for the expensive colour, costs 98 euros
- ▶ The variety constraint, on its own, does not say *how many* trees must use each configuration: one is enough
- ▶ It changes the right-hand side of the variety constraint, not the structure

Worked variant: all the configurations (2/2)

The variety constraint is not priced: the recipe of the base problem keeps $\gamma = \delta = 0$ and puts the whole price on a single colour, so the bound stays 2140.

	value	what it is
UB	3023	heuristic solution
LB	2140	dual certificate built by hand
$z(LP)$	2140	relaxation without the bounds
$z(LP^+)$	2190	relaxation with the bounds
$z(MILP)$	2239	optimum of the MILP

The heuristic installs the two missing configurations on one tree each and costs 3023: the certified gap stays wide and here the MILP really is needed.

Summary: Trees and lights

- ▶ Two storeys tied by an availability: no big-M, and the count of the boxes comes from the integrality of y_b
- ▶ The link $z_c \leq x_c$ is enough because x_c is a count; the missing direction is imposed by optimality
- ▶ Bounds: $UB = 3121$ (two-phase heuristic), $LB = 2140$ (only colour 2 is priced), $z(MILP) = 2141$
- ▶ The dual bound is off by a single unit, but the heuristic gap is 45.8% and the certified one 31.4%

Shipments in boxes

Introduction: Shipments in boxes

- ▶ We decide **how many units** to ship from every plant to every customer and **how many boxes** are needed on every route
- ▶ It is a multi-product flow in which the cost is not on the flow, but on the **containers**
- ▶ We will see how the capacity constraint, together with integrality, produces a **rounding up** without writing any ceiling
- ▶ We will build two bounds: a primal one from a constructive heuristic and a dual one, and then an integer bound stronger than the dual one

Problem 10.5: shipments in boxes (1/2)

- ▶ A company organises the shipments from its $n \in \mathbb{Z}_{\geq 1}$ production plants to its $m \in \mathbb{Z}_{\geq 1}$ customers
- ▶ The company produces $k \in \mathbb{Z}_{\geq 1}$ product types
- ▶ Products are shipped in **boxes**: every box travels directly from one plant to one customer
- ▶ Every box has maximum capacity $w \in \mathbb{Z}_{\geq 1}$, that is the maximum number of product units it can carry

Problem 10.5: shipments in boxes (2/2)

- ▶ For every product $p \in \{1, 2, \dots, k\}$ and every customer $c \in \{1, 2, \dots, m\}$, the value $d_{pc} \in \mathbb{Z}_{\geq 0}$ is the number of units ordered
- ▶ For every product p and every plant $s \in \{1, 2, \dots, n\}$, the value $a_{ps} \in \mathbb{Z}_{\geq 0}$ is the number of units available
- ▶ The company wants a shipment plan that **minimises the total number of boxes used**

The problem in words

- ▶ We **decide** how many units of each product to ship from each plant to each customer, and how many boxes are needed on each route
- ▶ **The objective:** the minimum number of boxes
- ▶ **The constraints:** every order met exactly, no plant beyond its stock, and enough boxes on every route for the units shipped

For brevity: $P = \{1, \dots, k\}$ the products, $S = \{1, \dots, n\}$ the plants, $C = \{1, \dots, m\}$ the customers.

Data (model input)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of plants, $s \in S$
m	$\in \mathbb{Z}_{\geq 1}$	number of customers, $c \in C$
k	$\in \mathbb{Z}_{\geq 1}$	number of products, $p \in P$
w	$\in \mathbb{Z}_{\geq 1}$	capacity of one box, in product units
d_{pc}	$\in \mathbb{Z}_{\geq 0}$	units of product p ordered by customer c
a_{ps}	$\in \mathbb{Z}_{\geq 0}$	units of product p available at plant s

Decision variables

$k \ n \ m$ non-negative integer variables for the flow:

x_{psc} = units of product p shipped from s to c

$n \ m$ non-negative integer variables for the boxes:

y_{sc} = boxes shipped from s to c

- ▶ No variable is binary: they are all **counts**
- ▶ The y_{sc} are the only ones appearing in the objective

Definition 4: Model 10.5 — shipments in boxes: objective and availability

$$\begin{aligned} \min \quad & \sum_{s=1}^n \sum_{c=1}^m y_{sc} \\ \text{s.t.} \quad & \sum_{s=1}^n x_{psc} = d_{pc}, \quad \forall p \in P, \forall c \in C \\ & \sum_{c=1}^m x_{psc} \leq a_{ps}, \quad \forall p \in P, \forall s \in S \end{aligned}$$

Definition 5: Model 10.5 — shipments in boxes: capacity and domains

$$-\sum_{p=1}^k x_{psc} + w y_{sc} \geq 0, \quad \forall s \in S, \forall c \in C$$

$$x_{psc} \in \mathbb{Z}_{\geq 0}, \quad \forall p \in P, \forall s \in S, \forall c \in C$$

$$y_{sc} \in \mathbb{Z}_{\geq 0}, \quad \forall s \in S, \forall c \in C$$

Model description

- ▶ The **objective** counts the boxes used on all the legs
- ▶ The **demand** constraints, one per product–customer pair, say that every order is met exactly ($k m$ equalities)
- ▶ The **availability** constraints, one per product–plant pair, do not allow shipping more than the plant has ($k n$ constraints)
- ▶ The **capacity** constraints, one per leg, say that the boxes shipped from s to c are enough to hold the units loaded ($n m$ constraints)

The link is a rounding up (1/2)

The capacity constraint reads

$$\sum_{p=1}^k x_{psc} \leq w y_{sc} \quad \Longleftrightarrow \quad y_{sc} \geq \frac{\sum_{p=1}^k x_{psc}}{w}$$

and since y_{sc} is a non-negative integer this is equivalent to

$$y_{sc} \geq \left\lceil \frac{\sum_{p=1}^k x_{psc}}{w} \right\rceil$$

- ▶ The ceiling never appears in the model: **integrality** produces it

The link is a rounding up (2/2)

The same constraint also imposes the implication

$$\sum_{p=1}^k x_{psc} > 0 \implies y_{sc} \geq 1$$

- ▶ If even a single unit leaves a plant towards a customer, at least one box must travel on that route
- ▶ The contrapositive is easier to read: if $y_{sc} = 0$ then $\sum_{p=1}^k x_{psc} = 0$
- ▶ The other direction — if nothing is shipped then no box — is not imposed by the constraint, but follows from **optimality**: every box costs 1 and the objective is a minimisation

The model in gurobipy (1/2)

```
m = gp.Model("shipments")
x = m.addVars(nk, nn, nm, vtype=GRB.INTEGER, name="x")
y = m.addVars(nn, nm, vtype=GRB.INTEGER, name="y")
m.setObjective(y.sum(), GRB.MINIMIZE)
```

- ▶ The x have three indices: product, plant, customer
- ▶ The objective sums all the y

The model in gurobipy (2/2)

```
m.addConstrs((x.sum(p, "*", c) == d[p][c]
               for p in range(nk) for c in range(nm)),
              name="demand")
m.addConstrs((x.sum(p, s, "*") <= a[p][s]
               for p in range(nk) for s in range(nn)),
              name="availability")
m.addConstrs((w * y[s, c] - x.sum("s, c") >= 0
               for s in range(nn) for c in range(nm)),
              name="capacity")
```

The instance

$n = 2$ plants, $m = 2$ customers, $k = 2$ products, $w = 10$:

d_{pc}	$c = 1$	$c = 2$
$p = 1$	5	0
$p = 2$	2	4

a_{ps}	$s = 1$	$s = 2$
$p = 1$	8	6
$p = 2$	5	7

The units to ship are $5 + 0 + 2 + 4 = 11$ in total.

Model 10.5 on the instance (1/2)

Eight variables x_{psc} for the flow and four y_{sc} for the boxes.

$$\min \quad y_{11} + y_{12} + y_{21} + y_{22}$$

$$\text{s.t.} \quad x_{111} + x_{121} = 5, \quad x_{112} + x_{122} = 0$$

$$x_{211} + x_{221} = 2, \quad x_{212} + x_{222} = 4$$

$$x_{111} + x_{112} \leq 8, \quad x_{121} + x_{122} \leq 6$$

$$x_{211} + x_{212} \leq 5, \quad x_{221} + x_{222} \leq 7$$

Model 10.5 on the instance (2/2)

$$-x_{111} - x_{211} + 10y_{11} \geq 0$$

$$-x_{112} - x_{212} + 10y_{12} \geq 0$$

$$-x_{121} - x_{221} + 10y_{21} \geq 0$$

$$-x_{122} - x_{222} + 10y_{22} \geq 0$$

$$x_{111}, x_{112}, x_{121}, x_{122}, x_{211}, x_{212}, x_{221}, x_{222} \in \mathbb{Z}_{\geq 0}$$

$$y_{11}, y_{12}, y_{21}, y_{22} \in \mathbb{Z}_{\geq 0}$$

Constructive heuristic: the primal bound (1/2)

- ▶ We proceed **customer by customer**: we try to serve each one from a single plant, the one that has everything they need
- ▶ If no plant is enough on its own, the order is split among several plants
- ▶ At the end one counts $\lceil \cdot / w \rceil$ boxes per route

It is a constructive heuristic of the family seen in the chapter on constructive heuristics: a single solution is built, one element at a time, never backtracking.

Constructive heuristic: the primal bound (2/2)

On the instance:

- ▶ **Customer 1** asks for 5 units of product 1 and 2 of product 2: plant 1 has 8 and 5, so it is enough on its own
- ▶ **Customer 2** asks for 4 units of product 2: after the first shipment plant 1 has 3 left, plant 2 has 7, so the shipment leaves from there
- ▶ There are two routes, with 7 and 4 units: one box each

$$z(MILP) \leq UB = 2$$

Dual of the LP relaxation of model 10.5

Dual variables: α_{pc} free on demand, $\beta_{ps} \leq 0$ on availability, $\gamma_{sc} \geq 0$ on capacity.

Definition 6: Dual of the LP relaxation of model 10.5

$$\begin{aligned} \max \quad & \sum_{p=1}^k \sum_{c=1}^m d_{pc} \alpha_{pc} + \sum_{p=1}^k \sum_{s=1}^n a_{ps} \beta_{ps} \\ \text{s.t.} \quad & \alpha_{pc} + \beta_{ps} - \gamma_{sc} \leq 0, & \forall p \in P, \forall s \in S, \forall c \in C \\ & w \gamma_{sc} \leq 1, & \forall s \in S, \forall c \in C \\ & \alpha_{pc} \geq 0, \quad \beta_{ps} \leq 0, \quad \gamma_{sc} \geq 0 \end{aligned}$$

The dual on the instance (1/2)

Four free α_{pc} , four non-positive β_{ps} , four non-negative γ_{sc} .

$$\max \quad 5\alpha_{11} + 2\alpha_{21} + 4\alpha_{22} + 8\beta_{11} + 6\beta_{12} + 5\beta_{21} + 7\beta_{22}$$

$$\text{s.t.} \quad \alpha_{11} + \beta_{11} - \gamma_{11} \leq 0, \quad \alpha_{12} + \beta_{11} - \gamma_{12} \leq 0$$

$$\alpha_{11} + \beta_{12} - \gamma_{21} \leq 0, \quad \alpha_{12} + \beta_{12} - \gamma_{22} \leq 0$$

$$\alpha_{21} + \beta_{21} - \gamma_{11} \leq 0, \quad \alpha_{22} + \beta_{21} - \gamma_{12} \leq 0$$

$$\alpha_{21} + \beta_{22} - \gamma_{21} \leq 0, \quad \alpha_{22} + \beta_{22} - \gamma_{22} \leq 0$$

The dual on the instance (2/2)

$$10\gamma_{11} \leq 1, \quad 10\gamma_{12} \leq 1$$

$$10\gamma_{21} \leq 1, \quad 10\gamma_{22} \leq 1$$

$$\alpha_{11}, \alpha_{12}, \alpha_{21}, \alpha_{22} \geq 0$$

$$\beta_{11}, \beta_{12}, \beta_{21}, \beta_{22} \leq 0$$

$$\gamma_{11}, \gamma_{12}, \gamma_{21}, \gamma_{22} \geq 0$$

- α_{12} does not appear in the objective because $d_{12} = 0$

Dual description

- ▶ α_{pc} is the value of one unit of product p delivered to customer c ; β_{ps} the (non-positive) price of the availability at plant s ; γ_{sc} the price of one unit of space on the leg from s to c
- ▶ **Columns of the x_{psc} :** shipping one unit is worth $\alpha_{pc} + \beta_{ps}$ and takes up one unit of space at price γ_{sc} ; the balance cannot be positive, because those variables do not appear in the objective
- ▶ **Columns of the y_{sc} :** one box offers w units of space, and their value cannot exceed 1, which is what the box costs

A dual solution by hand

- ▶ The constraint on the columns of the y_{sc} caps $\gamma_{sc} \leq 1/w$: take the maximum, $\bar{\gamma}_{sc} = 1/w$
- ▶ With $\bar{\beta} = 0$ the constraint on the columns of the x_{psc} becomes $\alpha_{pc} \leq 1/w$: take $\bar{\alpha}_{pc} = 1/w$

$$LB = \frac{1}{w} \sum_{p=1}^k \sum_{c=1}^m d_{pc} = \frac{11}{10}$$

- ▶ Every unit ordered takes $1/w$ of a box: it is the **volumetric** bound, and it coincides with $z(LP)$

A stronger integer bound

The volumetric bound ignores that a box cannot be split between two customers: every customer c with at least one unit ordered receives at least $\lceil \sum_p d_{pc}/w \rceil$ boxes, different from those of the other customers.

Customer	units ordered	minimum boxes
1	7	1
2	4	1
total	11	2

$$z(MILP) \geq LB = 2$$

almost twice the volumetric bound $11/10$.

Optimal solution

The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there.

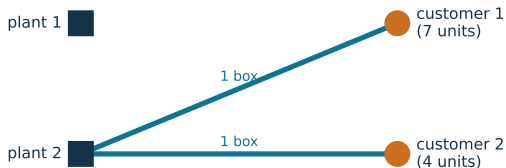
- ▶ Plant 2 serves both customers: customer 1 with one box holding 5 units of product 1 and 2 of product 2, customer 2 with one box holding 4 units of product 2

$$2 = LB \leq z(MILP) = 2 \leq UB = 2, \quad z(LP) = z(LP^+) = \frac{11}{10}$$

- ▶ The heuristic gap is zero and the integer bound certifies optimality
- ▶ The difference $2 - 11/10 = 9/10$ is entirely due to integrality: 45% of the optimal value

The optimal plan

10.5: optimal plan with 2 boxes



Which plants serve which customers, and with how many boxes.

Additional remarks (1/2)

- ▶ The model is a **multi-product flow** with costs on the containers only
- ▶ If the costs were on the flow (per unit carried) and there were no boxes, the problem would be a **pure transportation problem**: solvable in polynomial time and with an integral relaxation
- ▶ The variables x_{psc} could be continuous without changing the optimum, because the data are integer and the transportation matrix is totally unimodular *for fixed y*

Additional remarks (2/2)

- ▶ The x_{psc} stay integer because the problem speaks of **units of product**: a choice of fidelity to the statement, not of strength of the model
- ▶ The capacity constraint uses the same capacity w on every route
- ▶ With boxes of different sizes one would need several families $y_{sc}^{(t)}$, one per type, as in problem 10.4

Modelling question: separate products

Different products cannot travel in the same box.

How does the model change? What is the new optimum?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
UB	2	heuristic solution
LB	2	certified bound
$z(MILP)$	2	MILP optimum

- ▶ Gap of the heuristic: $\frac{|UB - z(MILP)|}{|z(MILP)|} = 0.0\%$
- ▶ Certified gap, known *before* solving the MILP (it cannot have $z(MILP)$ in the denominator): $UB - LB = 0$, that is $\frac{UB - LB}{|UB|} = 0.0\%$

Worked variant: smaller boxes (1/2)

The boxes hold 4 units instead of 10: only the datum w changes.

- ▶ Customer 1, who orders 7 units, now requires at least $\lceil 7/4 \rceil = 2$ boxes
- ▶ Customer 2 at least $\lceil 4/4 \rceil = 1$ box
- ▶ The integer bound becomes $LB = 2 + 1 = 3$ and is exact again; the heuristic uses 3, so the sandwich closes

Worked variant: smaller boxes (2/2)

The dual recipe stays $\beta = 0$ and $\gamma = \alpha = 1/w$: the bound of the relaxation is still units ordered divided by the capacity of the box, that is $11/4$ instead of $11/10$.

	value	what it is
UB	3	heuristic solution
LB	3	count of the boxes per customer
$z(LP)$	$\frac{11}{4}$	relaxation without the bounds
$z(LP^+)$	$\frac{11}{4}$	relaxation with the bounds
$z(MILP)$	3	MILP optimum

Summary: Shipments in boxes

- ▶ Multi-product flow with the cost on the **containers**: the constraint $\sum_p x_{psc} \leq w y_{sc}$ plus integrality is equivalent to a rounding up
- ▶ Constructive heuristic customer by customer: $UB = 2$
- ▶ Volumetric dual bound $LB = 11/10 = z(LP)$; the per-customer count raises it to $LB = 2$ and certifies the optimum $z(MILP) = 2$
- ▶ With boxes of 4 units the optimum rises to 3: the datum changes, not the structure of the certificate

Summer camps

Introduction: Summer camps

- ▶ We decide **how many children** of each nationality and of each sex to send to each camp
- ▶ The variables are not binary but **counts**: it is the first family in which they take large values
- ▶ We will see two camp composition rules and the difference between “not fewer than each other” and “not fewer than all the others together”
- ▶ We will build a primal bound with a constructive heuristic, a dual bound by hand and two combinatorial arguments

Problem 10.6: summer camps (1/2)

- ▶ A company runs $r \in \mathbb{Z}_{\geq 1}$ summer camps to host children during the holidays
- ▶ For every camp $j \in \{1, 2, \dots, r\}$, the value $d_j \in \mathbb{Z}_{\geq 1}$ is the maximum number of children that can be hosted
- ▶ The requests come from children of $s \in \mathbb{Z}_{\geq 1}$ different nationalities: for every nationality i there are $f_i \in \mathbb{Z}_{\geq 0}$ girls and $g_i \in \mathbb{Z}_{\geq 0}$ boys

Problem 10.6: summer camps (2/2)

- ▶ In every camp the number of **girls** must be greater than or equal to the number of boys
- ▶ In every camp the number of children of nationality $c \in \{1, 2, \dots, s\}$ must be greater than or equal to that of **every other** nationality
- ▶ The company wants to **maximise the total number of children accepted**

The problem in words

- ▶ We **decide** how many children of each nationality and of each sex to send to each camp
- ▶ **The objective:** the largest number of children accepted
- ▶ **The constraints:** neither the availabilities nor the capacities may be exceeded; and in every camp the two composition rules must hold

Data (model input)

Symbol	Type	Meaning
s	$\in \mathbb{Z}_{\geq 1}$	number of nationalities, $i \in \{1, 2, \dots, s\}$
r	$\in \mathbb{Z}_{\geq 1}$	number of camps, $j \in \{1, 2, \dots, r\}$
f_i	$\in \mathbb{Z}_{\geq 0}$	girls available of nationality i
g_i	$\in \mathbb{Z}_{\geq 0}$	boys available of nationality i
d_j	$\in \mathbb{Z}_{\geq 1}$	capacity of camp j
c	$\in \{1, 2, \dots, s\}$	nationality that must be the majority

Decision variables

The variables are not binary but *counts*: 2 *r s* non-negative integer variables.

x_{ij} = girls of nationality i in camp j

y_{ij} = boys of nationality i in camp j

- ▶ Every child accepted contributes one unit to the objective
- ▶ The places not assigned simply stay empty

Model 10.6 (1/2)

Definition 7: Model 10.6 — summer camps: objective, availability and capacity

$$\begin{aligned} \max \quad & \sum_{i=1}^s \sum_{j=1}^r (x_{ij} + y_{ij}) \\ \text{s.t.} \quad & \sum_{j=1}^r x_{ij} \leq f_i, & \forall i \in \{1, \dots, s\} \\ & \sum_{j=1}^r y_{ij} \leq g_i, & \forall i \in \{1, \dots, s\} \\ & \sum_{i=1}^s (x_{ij} + y_{ij}) \leq d_j, & \forall j \in \{1, \dots, r\} \end{aligned}$$

Definition 8: Model 10.6 — summer camps: composition and domains

$$\sum_{i=1}^s (x_{ij} - y_{ij}) \geq 0, \quad \forall j \in \{1, \dots, r\}$$

$$x_{cj} + y_{cj} - x_{ij} - y_{ij} \geq 0, \quad \forall i \neq c, \forall j \in \{1, \dots, r\}$$

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad y_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, \forall j$$

Model description

- ▶ The **availability** constraints cap the girls and the boys of each nationality ($2s$ constraints)
- ▶ The **capacity** constraints cap the places of each camp (r constraints)
- ▶ The **balance** constraints impose girls $\geq$ boys in every camp (r constraints)
- ▶ The **majority** constraints impose that nationality c is not fewer than each of the others, one inequality per nationality–camp pair $((s - 1) r$ constraints)

The majority constraint with more than two nationalities (1/2)

The majority constraint translates the statement literally: nationality c is not fewer than *each* of the others, $(s - 1) r$ inequalities in all. There is however a second, shorter writing:

$$x_{cj} + y_{cj} \geq \sum_{i \neq c} (x_{ij} + y_{ij}) \quad \forall j \in \{1, \dots, r\}$$

- ▶ These are r inequalities instead of $(s - 1) r$
- ▶ They say something else, and it is **stronger**: nationality c takes at least half of the places in every camp

The majority constraint with more than two nationalities (2/2)

- ▶ The two readings **coincide for** $s = 2$: with a single other nationality the sum has one term
- ▶ They diverge for $s > 2$, where the aggregated form cuts off solutions the statement allows
- ▶ The choice is made by reading the text, not for convenience of writing: fewer constraints does not mean a better model if they are not the constraints of the problem
- ▶ On the instance $s = 2$, so the two forms give the same numbers; and it is precisely because $s = 2$ that the combinatorial argument can read the majority as half of the places

The model in gurobipy (1/2)

```
m = gp.Model("camps")
x = m.addVars(s, r, vtype=GRB.INTEGER, name="x")
y = m.addVars(s, r, vtype=GRB.INTEGER, name="y")
m.setObjective(gp.quicksum(x[i, j] + y[i, j]
                           for i in range(s)
                           for j in range(r)),
               GRB.MAXIMIZE)
m.addConstrs((x.sum(i, "*") <= f[i] for i in range(s)),
              name="girls")
m.addConstrs((y.sum(i, "*") <= g[i] for i in range(s)),
              name="boys")
```

The model in gurobipy (2/2)

```
m.addConstrs((gp.quicksum(x[i, j] + y[i, j]
                           for i in range(s)) <= d[j]
               for j in range(r)), name="capacity")
m.addConstrs((gp.quicksum(x[i, j] - y[i, j]
                           for i in range(s)) >= 0
               for j in range(r)), name="balance")
m.addConstrs((x[c, j] + y[c, j] - x[i, j] - y[i, j] >= 0
               for i in range(s) if i != c
               for j in range(r)), name="majority")
```

The instance

$s = 2$ nationalities, $r = 2$ camps, $c = 1$:

	$i = 1$	$i = 2$
f_i	8	10
g_i	4	12

	$j = 1$	$j = 2$
d_j	15	8

In total there are 34 children available and 23 places.

Model 10.6 on the instance (1/2)

Four x_{ij} for the girls and four y_{ij} for the boys.

$$\max \quad x_{11} + x_{12} + x_{21} + x_{22} + y_{11} + y_{12} + y_{21} + y_{22}$$

$$\text{s.t.} \quad x_{11} + x_{12} \leq 8, \quad x_{21} + x_{22} \leq 10$$

$$y_{11} + y_{12} \leq 4, \quad y_{21} + y_{22} \leq 12$$

$$x_{11} + x_{21} + y_{11} + y_{21} \leq 15$$

$$x_{12} + x_{22} + y_{12} + y_{22} \leq 8$$

Model 10.6 on the instance (2/2)

$$x_{11} + x_{21} - y_{11} - y_{21} \geq 0$$

$$x_{12} + x_{22} - y_{12} - y_{22} \geq 0$$

$$x_{11} - x_{21} + y_{11} - y_{21} \geq 0$$

$$x_{12} - x_{22} + y_{12} - y_{22} \geq 0$$

$$x_{11}, x_{12}, x_{21}, x_{22} \in \mathbb{Z}_{\geq 0}$$

$$y_{11}, y_{12}, y_{21}, y_{22} \in \mathbb{Z}_{\geq 0}$$

- The first two rows are the balance, the other two the majority

Constructive heuristic: the primal bound (1/2)

The problem is a maximisation, so the heuristic gives a **lower** bound.

- ▶ One camp is filled at a time
- ▶ In every camp we take first the majority nationality (girls and then boys) and then the others
- ▶ We stop as soon as one of the three constraints would break

It is a constructive heuristic of the family seen in the chapter on constructive heuristics: a single solution, one element at a time, never backtracking.

Constructive heuristic: the primal bound (2/2)

On the instance:

- ▶ **Camp 1** (capacity 15): all 8 girls and all 4 boys of nationality 1, then 3 girls of nationality 2. The camp is full: 12 of nationality 1 against 3 of nationality 2, and 11 girls against 4 boys
- ▶ **Camp 2** (capacity 8): nationality 1 is exhausted, so any child of nationality 2 would violate the majority. The camp stays empty

$$z(MILP) \geq LB = 15$$

Dual of the LP relaxation of model 10.6 (1/2)

Dual variables: $\alpha_i, \beta_i, \gamma_j \geq 0$ on the three groups of $\leq$ constraints, and $\delta_j, \varepsilon_{ij} \geq 0$ on the two composition groups.

Definition 9: Dual of the LP relaxation of model 10.6 — objective and columns of c

$$\begin{aligned} \min \quad & \sum_{i=1}^s f_i \alpha_i + \sum_{i=1}^s g_i \beta_i + \sum_{j=1}^r d_j \gamma_j \\ \text{s.t.} \quad & \alpha_c + \gamma_j - \delta_j - \sum_{k \neq c} \varepsilon_{kj} \geq 1, \quad \forall j \in \{1, \dots, r\} \\ & \beta_c + \gamma_j + \delta_j - \sum_{k \neq c} \varepsilon_{kj} \geq 1, \quad \forall j \in \{1, \dots, r\} \end{aligned}$$

Dual of the LP relaxation of model 10.6 (2/2)

Definition 10: Dual of the LP relaxation of model 10.6 — other columns and domains

$$\alpha_i + \gamma_j - \delta_j + \varepsilon_{ij} \geq 1, \quad \forall i \neq c, \forall j \in \{1, \dots, r\}$$

$$\beta_i + \gamma_j + \delta_j + \varepsilon_{ij} \geq 1, \quad \forall i \neq c, \forall j \in \{1, \dots, r\}$$

$$\alpha_i \geq 0, \quad \beta_i \geq 0, \quad \forall i \in \{1, \dots, s\}$$

$$\gamma_j \geq 0, \quad \delta_j \geq 0, \quad \varepsilon_{ij} \geq 0, \quad \forall i \neq c, \forall j$$

The dual on the instance (1/2)

With $s = 2$ and $c = 1$ only two ε are left: ε_{21} and ε_{22} .

$$\min \quad 8\alpha_1 + 10\alpha_2 + 4\beta_1 + 12\beta_2 + 15\gamma_1 + 8\gamma_2$$

$$\text{s.t.} \quad \alpha_1 + \gamma_1 - \delta_1 - \varepsilon_{21} \geq 1$$

$$\beta_1 + \gamma_1 + \delta_1 - \varepsilon_{21} \geq 1$$

$$\alpha_1 + \gamma_2 - \delta_2 - \varepsilon_{22} \geq 1$$

$$\beta_1 + \gamma_2 + \delta_2 - \varepsilon_{22} \geq 1$$

The dual on the instance (2/2)

$$\alpha_2 + \gamma_1 - \delta_1 + \varepsilon_{21} \geq 1$$

$$\beta_2 + \gamma_1 + \delta_1 + \varepsilon_{21} \geq 1$$

$$\alpha_2 + \gamma_2 - \delta_2 + \varepsilon_{22} \geq 1$$

$$\beta_2 + \gamma_2 + \delta_2 + \varepsilon_{22} \geq 1$$

$$\alpha_1, \alpha_2, \beta_1, \beta_2 \geq 0$$

$$\gamma_1, \gamma_2, \delta_1, \delta_2 \geq 0$$

$$\varepsilon_{21}, \varepsilon_{22} \geq 0$$

Dual description (1/2)

- ▶ α_i is the price of a place for the girls of nationality i and β_i that for the boys
- ▶ γ_j is the price of a place in camp j , δ_j the price of the balance constraint
- ▶ ε_{ij} is the price of the comparison between nationality i and the majority one in camp j
- ▶ The **objective** prices the availability of each nationality and the capacity of each camp

Dual description (2/2)

The dual constraints are the columns of the primal, and they split into two blocks because nationality c appears in the comparisons differently from the others.

- ▶ **Columns of the x_{cj} :** accepting one girl of the majority nationality uses one place of her nationality and one of the camp, raises the balance by one unit and *loosens* every comparison of that camp — hence the sum with the minus sign
- ▶ **Columns of the x_{ij} with $i \neq c$:** that girl tightens one comparison only, her own
- ▶ In both cases the total value must cover the unit contributed to the primal objective; for the boys the same holds, with the sign of the balance reversed

A dual solution by hand

The simplest recipe prices only the capacity: $\alpha = \beta = \delta = \varepsilon = 0$ and $\gamma_j = 1$ for every camp.

- ▶ All the dual constraints become $\gamma_j \geq 1$ and are satisfied

$$UB = \sum_{j=1}^r d_j = 15 + 8 = 23$$

- ▶ The reading is obvious: every accepted child takes one place, so no more children can be accepted than there are places
- ▶ And it is also **optimal**: on the relaxation without the bounds $z(LP) = 23$

Two more combinatorial arguments

The bound 23 is not the only one that can be read off the data:

- ▶ nationality c takes at least half of the places of every camp, and there are $f_c + g_c = 12$ children of that nationality: at most $2 \cdot 12 = 24$ accepted
- ▶ the girls are at least half in every camp, and there are 18 girls: at most $2 \cdot 18 = 36$ accepted

Argument	upper bound
capacity of the camps	23
majority nationality	24
girls available	36

On this instance the capacity wins, but that is not a rule.

Optimal solution

The optimum is obtained by solving the model: the two bounds just built are what helps the solver get there.

	camp 1		camp 2	
	girls	boys	girls	boys
nationality 1	8	0	0	4
nationality 2	1	6	4	0
total	15 out of 15		8 out of 8	

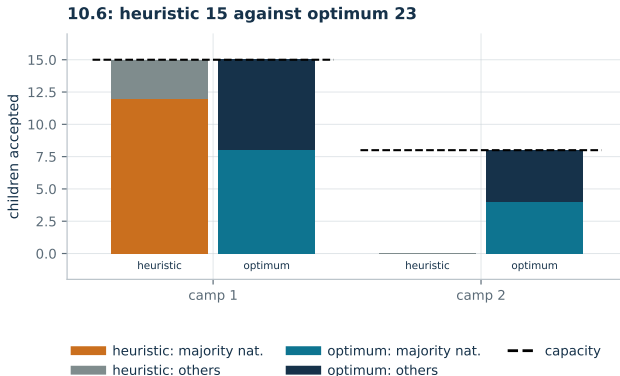
Both camps are full and the chain of bounds is

$$15 = LB \leq z(MILP) = 23 = z(LP) = UB$$

Optimal solution: where the heuristic goes wrong

- ▶ The heuristic gap is 34.8%, and the dual bound settles the problem
- ▶ The whole gap was on the **solution** side, not on the certificate side
- ▶ The mistake is clear: the heuristic uses up the majority nationality in the first camp, and in the second one nobody is left who can form the majority

Optimal solution: the two camps



Additional remarks

- ▶ The variables are integer but not binary: the counts take large values, and the relaxation stays tight anyway because all the constraints are sums
- ▶ The balance constraint and the majority constraint are **independent**: one can have camps with many girls and few children of nationality c , and vice versa. On the instance it is the second one that bites
- ▶ If a nationality had zero children available, the majority would automatically exclude it from every camp in which anyone else appears: a limit case worth checking on the data

Modelling question: a nationality that cannot be split

For organisational reasons the children of nationality 1 must all stay in the same camp.

How does the model change? What is the new optimum?

The final comparison

The three numbers that every problem of the course produces:

	value	what it is
LB	15	heuristic solution
UB	23	certified bound
$z(MILP)$	23	MILP optimum

- ▶ Gap of the heuristic: $\frac{|LB - z(MILP)|}{|z(MILP)|} = 34.8\%$
- ▶ Certified gap, known *before* solving the MILP (it cannot have $z(MILP)$ in the denominator): $UB - LB = 8$, that is $\frac{UB - LB}{|LB|} = 53.3\%$

Worked variant: a bigger camp (1/2)

Camp 1 is enlarged and reaches 20 places: only the datum d_1 changes.

- ▶ The total capacity rises to 28, but the optimum stops at 24
- ▶ The second combinatorial argument is now in charge, the one about the majority nationality: $2 \cdot (f_1 + g_1) = 2 \cdot 12 = 24$
- ▶ It is a good example of the fact that the right bound changes with the instance: the dual recipe $\gamma_j = 1$ would give 28, which is worse

Worked variant: a bigger camp (2/2)

The certificate does not change shape — $\gamma_j = 1$ and all the other dual variables at zero — but the datum it adds up does: $\sum_j d_j$ is now 28. The heuristic fills the camps in order and reaches 20.

	value	what it is
UB	28	dual certificate built by hand
LB	20	heuristic solution
$z(LP)$	24	relaxation without the bounds
$z(LP^+)$	24	relaxation with the bounds
$z(MILP)$	24	MILP optimum

Summary: Summer camps

- ▶ Integer **count** variables, availabilities, capacities and two composition rules
- ▶ “Not fewer than each other” and “not fewer than all the others together” coincide only for $s = 2$: the second is stronger
- ▶ Bounds: $LB = 15$ (heuristic), $UB = z(LP) = 23$ (dual recipe $\gamma_j = 1$), $z(MILP) = 23$; heuristic gap 34.8%
- ▶ With camp 1 at 20 places the capacity certificate gives 28, but the majority nationality argument is in charge: $z(MILP) = 24$